

Argument	Description
formula	symbolic description of the model
data	data frame, list or environment
subset	optional vector of observations
weights	optional vector of weights
method	only 'qr' is supported
singular.ok	logical value, error if false
contrasts	optional list
offset	<i>a priori</i> known component
digits	significant digits

A more detailed output on the relation 'r' can be printed using the `summary()` function, as shown below:

```
> summary(r)
```

```
Call:
lm(formula = y ~ x)
```

```
Residuals:
    1      2      3      4      5 
4.367e-16 -8.276e-16  3.028e-16  1.303e-16 -4.222e-17
```

```
Coefficients:
            Estimate Std. Error  t value Pr(>|t|)
(Intercept) 1.000e+00  5.207e-16  1.920e+15  <2e-16 ***
x            1.000e+00  9.065e-17  1.103e+16  <2e-16 ***
---

```

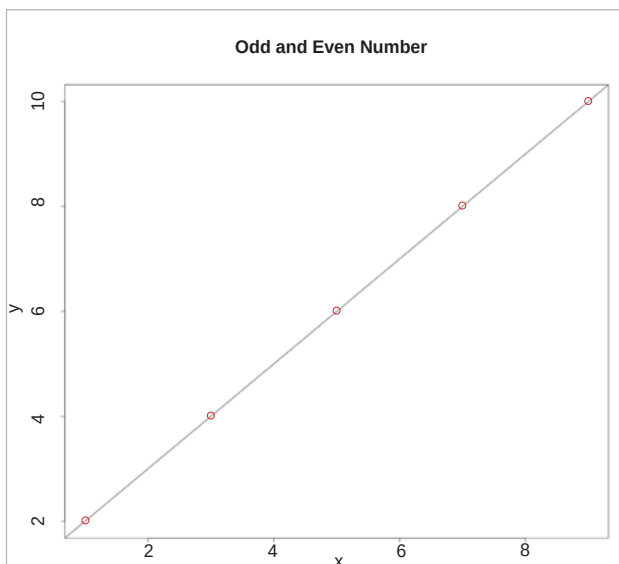


Figure 1: Plot x and y

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5.733e-16 on 3 degrees of freedom
Multiple R-squared:  1, Adjusted R-squared:  1
F-statistic: 1.217e+32 on 1 and 3 DF, p-value: < 2.2e-16
```

We can use the `predict()` function to find the possible value of 'y' for a given 'x'. For example:

```
> v <- data.frame(x = 11)
> p <- predict(r, v)
> p
    1
   12
```

The `plot()` function is used to display the fitted model, as illustrated below:

```
> plot(y, x, col="red", main = "Odd and Even Numbers",
       abline(lm(x~y)))
```

The normal QQ plot for the residual values can be displayed using the `qqnorm()` and `qqline()` functions, as shown below:

```
> qqnorm(r$resid)
> qqline(r$resid)
```

The kernel density estimates on the residuals can also be graphed using the `plot()` function, as shown in the figure.

```
> plot(density(r$resid))
```

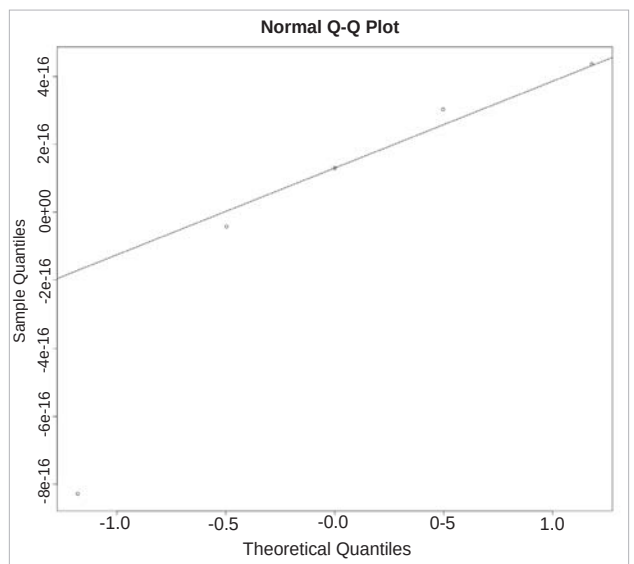


Figure 2: QQ plot